



Instructor: Date:

Learner: Regional Branch:

Program: Center:

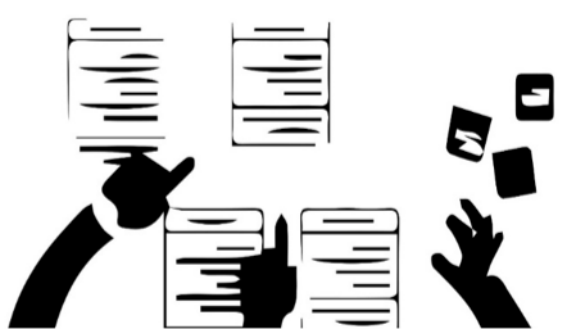
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Let's Start!

Components of Object Oriented Software System



Learning Outcome:

Describe common simple incidents at work in a simple spoken or written situation with a supportive interlocutor. Build the macro system conceptual model following the customer requirements, through the use and interpretation of collected information, represented in class diagrams, interaction diagrams, and operation contracts.

Can Do Statements:

- I can say what a UML class diagram shows.
- I can talk about the elements that appear in a class.
- I can write a simple text on how to draw a basic class diagram.

Ask your partner:

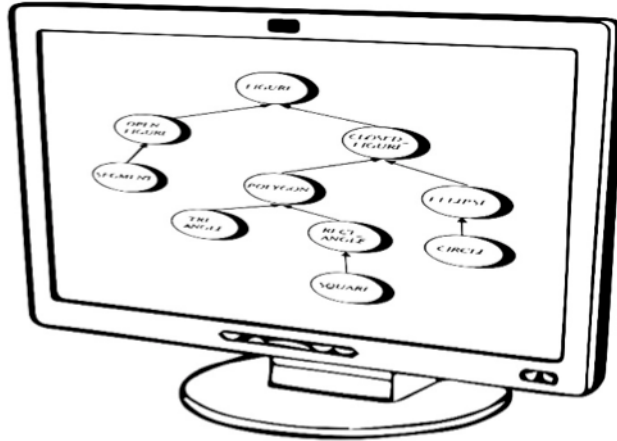
- What kind of information can you find in a diagram? Give examples.
- What kinds of diagrams do you know? What are they for?



Let's Explore!

A system engineer gives a talk about UML* class diagrams.

*"UML" means Unified Modelling Language.



Instructions: Read the text below.

Professor Smith: A class is a design that describes the components of a software system. You need to use a class diagram to represent any kind of information that you want to organize. A UML class diagram shows multiple classes in detail and relates them to each other. A class diagram also shows the software architecture of a system, and you need to represent that architecture in a diagram.

If you need to draw a class diagram, you don't need to make all information visible. A UML allows us to suppress any idea that we consider irrelevant, so the diagram can be clear and understandable. You need to clarify the class that you want to organize, but you don't need to organize it before creating a box structure. The box structure doesn't need to be complex, but it has some specifications that you have to follow, like the elements of a class diagram. These elements are first, the class name that has to be unique. Second, the list of attributes that you need to describe the class. And third, a list of methods or operations that you need to follow for a UML class.

Attributes and methods use some modifiers and diagram conventions that differentiate one from another. A class diagram uses (+) for public access and (-) for private access. Attributes are normally private and methods are normally public. You need to specify the type of visibility you want for your diagram, but you don't need to use all of them at the same time. Don't forget that you can use keywords to create your class diagram. Now, it's your turn, according to what you learned today, you need to write a short text explaining how to draw a basic class diagram. You can start right now!



Let's Practice!

What does a class diagram show?

Instructions: Use the words from the word bank to complete each sentence.

Word Bank

illustrate / illustrates / illustrating / shows / shown / showing /
consist / consists / consisting

1. A UML class diagram _____ multiple classes in detail.
2. A class diagram _____ the software architecture of a system.
3. The diagram _____ in the representation of that architecture.



Let's Practice!

What are the elements of a class diagram?

Instructions: Organize the mystery sentences.

1. follow / . / You / class / a / to / of / need / list / methods / UML / operations / for / or

2. The / unique. / class / to / be / needs / name / but / it / need / be / to / in /
English, / doesn't

3. class / . / UML / a / for / list / of / to / follow / methods / or / You / need / operations



Extension Activity

Designing a Class Diagram

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more specificities about class diagram design.
- Design a class diagram for an object. The object could be a person, a thing or a software.
- Explain your diagram using the grammar related to this lesson.
- Share your class diagram with your partners and instructor.

Self-Assessment

Yes **No**

- | | | |
|---|-----------------------|-----------------------|
| 1. I research about class diagram design. | ☐ | ☐ |
| 2. I design a class diagram for an object. | ☐ | ☐ |
| 3. I use the grammar related to this lesson to explain my diagram design. | ☐ | ☐ |
| 4. I share my class diagram with my partners and teacher. | ☐ | ☐ |



Target Words

- **software architecture** (noun): the software system's structure
e.g. Class diagram also shows the software architecture of a system and you need to represent that architecture in a diagram.
- **suppress** (verb): to exclude
e.g. UML allows us to suppress any idea we consider irrelevant, so the diagram can be clear and understandable.
- **diagram convention** (noun): a notation
e.g. Attributes and methods use some modifiers and diagram conventions that differentiate one from another.
- **box structure** (noun): a diagram you use to represent a class
e.g. You need to clarify the class you want to organize but you don't need to organize it before creating a box structure.
- **attribute** (noun): a characteristic
e.g. Attributes are normally private and methods are normally public.
- **diagram** (noun): a graphic representation of any information
e.g. You need to use a class diagram to represent any kind of information you want to organize.



Learning Points

Simple Present Tense in Affirmative Statements

You use **simple present tense in affirmative statements** to describe actions that are true, normal or regular.

You can use the simple present tense in affirmative statements to talk about real facts.

You just need to follow this grammatical structure:

Subject + ***Verb** + Complement

Example:

*Class diagrams **organize** many different types of information.*

*Don't forget the right conjugation of verbs for the third persons of the singular.

Subject	Verb	Complement
<i>I – You – We – They</i>	<i>describe</i>	<i>the diagram.</i>
<i>He – She – It</i>	<i>describes</i>	<i>the diagram.</i>

Examples

- *A class is a design that describes the components of a software system.*
- *UML allows us to suppress any idea we consider irrelevant so the diagram can be clear and understandable.*
- *Class diagram uses (+) for public access and (-) for private access.*





Learning Points

Verbs with Infinitives, Need to, in Affirmative and Negative Statements

You use **verb + infinitive in affirmative and negative statements with 'need to'**, to express necessity for something.

You use the following structure when writing a verb + infinitive in affirmative statements with "need to".

Subject + **Need** + **to** + Main verb in base form + Complement

Example:

*I **need to** describe a class diagram.*

You can also use verbs with infinitives in negative statements by adding the auxiliaries don't or doesn't respectively.

Subject + **don't /doesn't** + **Need** + **to** + Main verb in base form + Complement

Examples:

*I **don't need to** describe a class diagram.*

*He **doesn't need to** describe a class diagram.*

Other kinds of verbs + infinitives are have to and want to.

Examples

- *You need to use a class diagram to represent any kind of information you want to organize.*
- *If you need to draw a class diagram you don't need to make all information visible.*
- *You need to specify the type of visibility you want for your diagram, but you don't need to use all of them at the same time.*
- *The box structure doesn't need to be complex, but it has some specification you have to follow, such as the elements of a class diagram.*



Learning Points

Drawing a Basic Class Diagram

Basic class diagrams are useful to represent specific components within one same category.

When drawing a basic class diagram, UML allows us to suppress irrelevant information in diagrams.

You need to choose an attribute of the category you want to work on. For example, a Class Software with design attributes and with some methods like life cycle, programming language, network protocols, and software samples.

Don't forget that you can use some diagram conventions to clarify your diagram:

- + Public
- # Protected
- Private

